The DeepSeek-R1 paper, mapped

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Paper: *DeepSeek-R1: Incentivizing Reasoning Capability in LLMs via Reinforcement Learning*, DeepSeek-AI, January 2025 (arXiv 2501.12948). **One-line summary:** a model can learn to reason step by step from reinforcement learning with simple, automatically checkable rewards, and that reasoning can then be copied into much smaller models.

The two models in the paper

	R1-Zero	R1
Starts from	DeepSeek-V3-Base (no instruction tuning)	DeepSeek-V3-Base
Supervised examples first?	None. Pure reinforcement learning	A small "cold start" set of thousands of long, readable worked solutions
RL algorithm	GRPO	GRPO
Rewards	Accuracy (is the final answer right?) + format (is the thinking inside `<think>` tags?)	Same, plus a language consistency reward; later a general stage that also uses reward models
Strength	Shows reasoning can emerge from RL alone	Reasons well **and** reads well
Weakness	Hard to read, mixes languages	More stages, more engineering

The R1 recipe, in four stages

Stage	What happens	Why
1. Cold start	Fine-tune on thousands of long chain-of-thought examples	Gives RL a readable starting style
2. Reasoning RL	GRPO on maths, code and logic with rule-based rewards + language consistency	Builds the reasoning skill
3. Rejection sampling + SFT	Keep good answers from the RL model: about 600k reasoning + 200k general samples, fine-tune for two epochs	Locks in good behavior, adds general skills
4. RL for all scenarios	Rule-based rewards for reasoning, reward models for helpfulness and harmlessness	Polishes the final assistant

Results worth remembering

Claim	Number (as reported in the paper)
R1-Zero, AIME 2024 pass@1, during RL	15.6% → 71.0%
R1-Zero with majority voting	86.7%
Qwen-32B: RL directly vs distilled from R1, AIME 2024	47.0 vs 72.6
Qwen-32B: RL directly vs distilled, MATH-500	91.6 vs 94.3

Words the paper uses

Term	Plain meaning
GRPO	Group Relative Policy Optimization (*DeepSeekMath*, Shao et al., 2024). Sample a group of answers to one question, score them, and push up the ones that beat the group average. No separate critic model.
Rule-based reward	A score computed by a simple program (compare the final number, run unit tests), not by another AI.
Aha moment	An intermediate R1-Zero checkpoint started writing things like "wait" and re-checking its own work, without being taught to.
Response length growth	During RL, R1-Zero's answers got steadily longer: it learned to spend more thinking on harder problems.
Distillation	Fine-tune a small "student" model on answers written by a big "teacher" model.
Cold start	A small supervised warm-up before RL.

What did not work (Section "Unsuccessful attempts")

- **Process reward models** (grading every step): hard to define a "step", easy to hack, costly to retrain.
- **Monte Carlo Tree Search** (searching over steps like a chess engine): the space of possible text is far too large.

Why this course cares

1. Rule-based rewards are cheap and hard to fool, so **you can run this on a free GPU** with GSM8K.
2. GRPO needs no critic model, which **saves memory**.
3. The paper's own finding, that distillation beats RL on small models, is **why Module 8 exists**.

Numbers are as reported by DeepSeek-AI (2025). They come from very large models and very large compute budgets; your small-model runs will look different.